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ORIGINAL PAPER

Unprecedented rainfall in the United Arab Emirates: hydrologic and flood impact analysis of the April 2024 event

Khalid Hussein[1,2] · Naeema Alhosani[1] · Ahmed M. Al-Areeq[3,4] · Amran A. Al Aghbari[5] · Muhagir Elkamali[1] · Tareefa Alsumaiti[1] · Hatim O. Sharif[6] · Ahmed M. G. Almurshidi[1] · Waleed Abdalati[2]

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Abstract
In mid-April 2024, an extreme rainfall event struck Oman and the United Arab Emirates (UAE), causing unprecedented flooding, significant infrastructure damage, and loss of life. Characterized by intense and sustained rainfall over a few days, this event highlighted the region's vulnerability to extreme weather phenomena. This study examines the rainfall estimated by two satellite rainfall products during this event, with a detailed analysis focusing on the watershed encompassing the city of Al Ain. A rain gauge in this watershed recorded a historic 24-hour rainfall of 254.8 mm, approximately 75% of the area's previously estimated probable maximum precipitation (PMP). Other gauges in the watershed also recorded substantial rainfall amounts, breaking previous records, while satellite products significantly underestimated the actual rainfall. A physically-based distributed hydrologic model simulated the resulting flood, indicating a low runoff ratio of about 7.14% due to high infiltration rates. Despite this, significant flooding occurred in urbanized parts of the watershed. This discrepancy highlights the complexity of urban hydrology and the challenges of predicting flood extents in urban areas. The findings underscore the need for improved flood forecasting systems in the UAE, emphasizing enhanced satellite rainfall estimation accuracy and advanced modeling approaches for better urban flood management and mitigation. Integrating new technologies and methodologies in urban flood forecasting is crucial for enhancing resilience against future extreme weather events, ensuring the safety and preparedness of vulnerable regions like the UAE.

Keywords Extreme rainfall · Flooding · Urban flood management · Arabian peninsula · UAE · Hydrologic modeling · Satellite rainfall products

1 Introduction

Severe weather events in the United Arab Emirates (UAE), particularly intense rainstorms, pose a significant challenge for public policy and climate change adaptation efforts (LTS 2024). Understanding the relationship between severe rainfall events and flooding is cru-

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cial for effective mitigation and adaptation strategies, such as managed aquifer recharge enhancement, especially given the scarcity of renewable water resources (Al Abdouli et al. 2019; Hussein et al. 2021). Floods can cause diverse economic impacts, including direct asset damage, business interruptions, and disruptions to global trade networks (Deryugina et al. 2018). Accurate meteorological data from climate models is pivotal for estimating these flood losses (Taguchi et al. 2022), and establishing robust flood warning systems is imperative to mitigate future events (Paul and Sharif 2018).

Severe weather impacts are a focal point in UAE public discourse and policy agendas (Salam 2015), particularly concerning the monitoring of infrequent rainfall events due to water scarcity (Al Abdouli et al. 2019). Urbanization significantly increases the susceptibility of cities to flooding (World Bank 2021). Studying weather extremes and their spatial patterns in urban environments is crucial for climate research (IPCC 2021). In the arid landscape of the UAE, sporadic heavy rainstorms occur unpredictably and with varying intensity (NCM, 2023), emphasizing the necessity for a comprehensive analysis of their attributes, impacts, and historical context. Such an examination is crucial for understanding the nation's vulnerability to extreme rainfall events, especially in urban settings. Moreover, various factors, including demographic changes, economic influences, and capital efficiency, significantly contribute to economic losses associated with floods. Incorporating these factors emerges as an effective strategy to enhance flood mitigation policies (Tanoue et al. 2020; Zhang et al., 2022).

In line with the need for such analyses, recent shifts in regional rainfall patterns across the Arabian Peninsula further underscore the vulnerability of the UAE to extreme weather events. While winter rainfall has increased in the eastern region, it has decreased in the south and northeast (Terry et al. 2023). This region, characterized by low annual rainfall and high variability, is becoming increasingly vulnerable to extreme flooding events (Dasari et al. 2021). Global climate change is intensifying extreme rainfall events, particularly in regions such as the Arabian Peninsula. Rising temperatures are amplifying rainfall intensity, which, in turn, heightens flood risks. (Terry et al. 2023; Hettiarachchi et al. 2018). In the UAE, storms are influenced by factors such as humid air masses, warm ocean temperatures, and mountain ranges (Patlakas et al. 2021; Abushandi and Al Ajmi 2022). These factors, along with recent shifts in rainfall patterns and intensity, have made the mountainous eastern region particularly vulnerable to flash floods (Merabtene et al. 2016; Chowdhury et al., 2016).

Given these shifting patterns and the increasing flood risks, the accuracy of rainfall data, especially at local scales, is essential for effective flood management. Numerous hydrometeorological studies in the UAE have systematically assessed the quality and hydrologic utility of satellite rainfall products (e.g., Alsumaiti et al. 2020; Almheiri et al. 2023; Kagone et al. 2023; Rachidi et al. 2023; Alsumaiti et al. 2024). These investigations have concentrated on the evaluation of diverse satellite precipitation datasets, employing rigorous statistical measures such as correlation coefficients, bias, mean square error, and efficiency coefficients. The results showed that satellite-based precipitation estimates agreed reasonably with rain gauge observations on a monthly and yearly basis, but the accuracy at shorter time scales was somewhat lower. (e.g., Alhosani et al. 2024). Notably, these studies emphasized the critical role of bias correction in enhancing the accuracy of regional precipitation products. By incorporating ground observations, researchers can effectively adjust these

products, resulting in more reliable hydrological and flood analysis models that directly rely on these rainfall datasets.

The investigation of rare weather extremes, their spatial variability, and their impacts on arid environments remains a significant challenge in socio-hydrological research. In the UAE, the unpredictable and intense nature of occasional rainfall necessitates a closer examination of their characteristics, causes, and consequences. This research aims to identify flood hazards, improve public preparedness, develop better flood mitigation strategies, and provide insights to disaster response agencies for more effective emergency interventions. By deepening our understanding of the UAE's vulnerability to extreme rainfall, we can enhance public awareness and support disaster agencies in planning more effective responses.

The study focuses on the unprecedented April 2024 rainfall event that set new records in the UAE, particularly in and around Al Ain city. Our investigation has four main objectives: (1) to describe the timeline and main features of the April 2024 extreme rainfall event; (2) to examine the event's rainfall characteristics using rain gauge and satellite data; (3) to utilize a physically based, distributed hydrologic model to simulate the severity of flooding across a watershed covering Al Ain City; and (4) to discuss changes in flood risk in the UAE and potential approaches for mitigation and adaptation.

1.1 The diverse landscapes of the UAE

The UAE is located on the Arabian Peninsula, sharing borders with Oman, Saudi Arabia, and the Arabian Gulf. The country's topography showcases a contrast between the eastern Oman Mountains, featuring Jebel Jais as the highest peak, and expansive low-lying plains that gradually transform into vast deserts in the western areas (Oxford Analytica, 2023). Despite being renowned for tourism growth, particularly in Dubai, the UAE also possesses rich geodiversity and biodiversity, offering potential for nature-based tourism experiences, including geo-tourism (Allan, 2023).

The climate of the UAE is characterized by a desert environment with two main seasons, winter and summer, separated by transitional periods. Winter (December to March) averages temperatures of 16.4 °C to 24 °C. The first transition period (April to May) sees temperatures gradually rising to 26 °C to 33.5 °C. Summer (June to September) is extremely hot, with temperatures reaching up to 50 °C in the south and averaging 32 °C to 37.2 °C. Rainfall is minimal, especially along the coasts. The second transition period (October to November) has milder temperatures of 24 °C to 30 °C. Rainfall is sparse throughout the UAE, averaging 140–200 mm annually. However, mountainous areas can receive up to 350 mm per year, leading to flash floods in temporary streams (wadis) (World Bank 2024). Dust storms and occasional snowfall in mountainous areas contribute to the dynamic landscape (Salameh et al. 2023). Often, hot wind exacerbates summer discomfort, while high-pressure systems over the Arabian Peninsula can enhance wind speeds and moisture inflow, influencing precipitation patterns (Gopalakrishnan et al. 2023). Urban planning studies in the UAE emphasize the impact of building heights on microclimates, potentially intensifying storms and increasing the risk of severe flooding (Nelli et al. 2022; Zhang et al. 2018; Mohammed Kishta et al. 2022).

The UAE faces severe water scarcity due to its arid climate and limited freshwater resources, making it one of the most water-stressed countries globally (Al-Karaghouli

and Kazmerski 2013; World Bank 2020). To address this challenge, the country heavily relies on desalination plants to meet its water demands (Alshaali et al., 2022). Groundwater sources are available but require careful management to prevent depletion. Runoff from sporadic storms is expected to increase significantly due to rapid urbanization in the coastal regions of the UAE, which presents opportunities to enhance water supplies (Al Abdouli et al. 2019). In addition to augmenting the water supply during droughts, the application of approaches such as Managed Aquifer Recharge (MAR) techniques in the UAE offers several potential benefits, including flood control, reduction of aquifer salinity, ecosystem rehabilitation, and the purification and management of runoff. (Ahmed et al. 2022; Ben Hassen 2022; Al Abdouli et al. 2019).

1.2 The 2024 extreme event

Despite its reputation for hot and dry weather, the region has experienced an increase in heavy rain events leading to floods in recent years (Oxford Analytica 2024). Experts attribute this increase to a combination of factors, including land use/land cover change and climate anomalies (Alhosani et al. 2024). In April 2024, heavy rains swept across extensive areas of the Arabian Peninsula, sparking devastating flash floods throughout the region. Some areas received nearly a year's worth of rain in just one day, resulting in widespread flooding and significant damage in Oman and the UAE (Singh 2024). During this event, the UAE encountered an unprecedented deluge, marking the most substantial rainfall event ever recorded in this desert nation. Heavy rains caused floods that primarily affected Dubai, Sharjah, Ajman, Fujairah, Ras Al Khaimah, and Umm Al Quwain (Bangar 2024). The rarity of rainfall in the UAE underscores the region's vulnerability to such extreme weather events.

The Royal Meteorological Society attributes the recent event to a mesoscale convective system (RMetS, 2024), while the UAE's National Centre of Meteorology cites a "pressure squeeze" caused by a low-pressure system in the upper atmosphere interacting with low pressure at the surface. This squeeze, fueled by warm ground temperatures and cooler air above, created ideal conditions for powerful thunderstorms (Reuters, 2024). Other contributing factors include a train of low-pressure systems moving along the jet stream towards the Arabian Gulf, as suggested by climate scientist Michael Mann, and the presence of large thunderstorms confirmed by meteorologists from the University of Reading (Knapton 2024). Nelli et al. (2022) found that all significant heavy rainfall events in the southern Arabian Peninsula coincided with El Niño years. This supports earlier findings by Kumar and Ourda (2014), who identified strong statistical links between precipitation patterns in the UAE and sea surface temperatures (SSTs) in both the equatorial Pacific and North Atlantic.

This rainfall event, characterized by historic intensity, inundated Dubai International Airport, causing extensive disruptions to flights at the world's busiest international airport. With standing water impeding taxiways and halting arrivals, the infrastructure struggled to cope, leaving passengers stranded on waterlogged roads (Fig. 1). Beyond aviation, the disruption led to school closures and remote work arrangements for government employees across the UAE. Severe disruptions affected Dubai's transportation network, leaving approximately 200 commuters stranded at metro stations due to service issues. Traffic flow was significantly impacted, with a major highway reduced to a single lane and the E11 route to Abu Dhabi completely closed in that direction (Desk 2024). Intercity bus services were



Fig. 1 Floodwaters turn Dubai airport tarmac into a lake (Sky News, 2024)

suspended, affecting travel between Dubai and neighboring emirates Sharjah and Ajman (Paul and Dhanjal 2024). Tragically, these conditions led to five confirmed deaths: three expatriates, whose causes of death were not determined, an elderly Emirati man, and an expatriate who drowned when the tanker he was driving swept into a wadi (Aghaddir, 2024).

The state-operated Wakalat Anba'a al-Emarat, WAM (Emirates News Agency), labeled the rainfall on April 16 and 17 as a "historic weather event," surpassing all documented records since 1949, predating the discovery of crude oil in the region when it was known as the Trucial States under British protectorate (Guardian News and Media 2024). Neighboring Bahrain, Oman, Qatar, and Saudi Arabia also experienced rainfall, but the UAE experienced a particularly pronounced impact. Starting late Monday (15/04/2024), the rainfall saturated Dubai with approximately 20 mm (0.79 inches), intensified through Tuesday morning, and totaled over 142 mm (5.59 inches) within 24 h—well above the annual average of 94.7 mm (3.73 inches) at Dubai International Airport (Press 2024). Authorities dealt with the aftermath as Khatm al-Shakla station near Al Ain city in Abu Dhabi Emirate recorded the highest rainfall accumulation of 254.8 mm (10 inches). Residents' images and video footage vividly illustrated the extent of the flooding in Al Ain City (Fig. 2).

2 Data and methods

2.1 Rainfall data

This study investigates the extraordinary storm event using both satellite rainfall estimates and rain gauge observations. The Integrated Multi-Satellite Retrievals for the Global Precipitation Mission (IMERG) is a highly promising advancement in multi-satellite precipitation products. Using IMERG data to simulate flooding in the UAE with a physically based model is beneficial because it provides high-resolution, near-real-time precipitation data, which



Fig. 2 Flooding on one of the streets of Al Ain city. Courtesy of Khalid Hussein

can potentially be used for flood forecasting. This data is particularly valuable in regions with sparse ground-based measurements, as it enhances the reliability of flood simulations and early warning systems, improving hydrological modeling and predictions. (Sharif et al. 2017). IMERG offers three options: Early (available within 6 h), Late (available within 18 h), and Final (available after 3 months). Early and Late products use climatological calibrations, while the Final product incorporates satellite-gauge data for improved accuracy. All products boast a fine spatial and temporal resolution of 0.1 degrees and 30 min. Studies comparing IMERG with earlier satellite products demonstrate a significant improvement in rainfall intensity estimation (Toh et al. 2023). Additionally, evaluations using ground-based rainfall data observed good agreement between IMERG estimates and ground observations for certain events (Mandapaka and Lo 2020). IMERG data for the April 15–17, 2024, storm event over the UAE was acquired from the NASA Precipitation Measurement Missions (PMM) website (https://gpm.nasa.gov/data).

To conduct a detailed analysis of flood impacts across the watershed, we utilized high-resolution rain gauge data (15-minute intervals, 0.2 mm minimum detection) from the UAE NCM network. This dataset includes data from nine rain gauges situated within and near the watershed, including ones that recorded the highest rainfall during the event. However, the complex terrain of the Oman Mountains restricts rain gauge placement, with most stations situated in towns and some stations located in neighboring Oman, limiting the ability to capture detailed rainfall data across the entire study area.

2.2 Al Ain watershed

The vast desert plains of Abu Dhabi Emirate, which also include a coastal region, host the city of Al Ain. This city stands in stark contrast to its surroundings, renowned for its green oases (springs) and the imposing Jebel Hafeet Mountain, a geological outlier of the Oman Mountains that shapes the local landscape. To assess the flooding response of the April

2024 extreme event, the Gridded Surface Subsurface Hydrologic Analysis (GSSHA) model was employed to the partially urbanized watershed spanning most of Al Ain City and its surrounding villages, covering an area of 2216 km² (Fig. 3). This watershed experienced the highest recorded rainfall during the event. Unlike most cities, which are built along the coast, Al Ain's inland location near the Oman Mountains provides it with a unique geographical context.

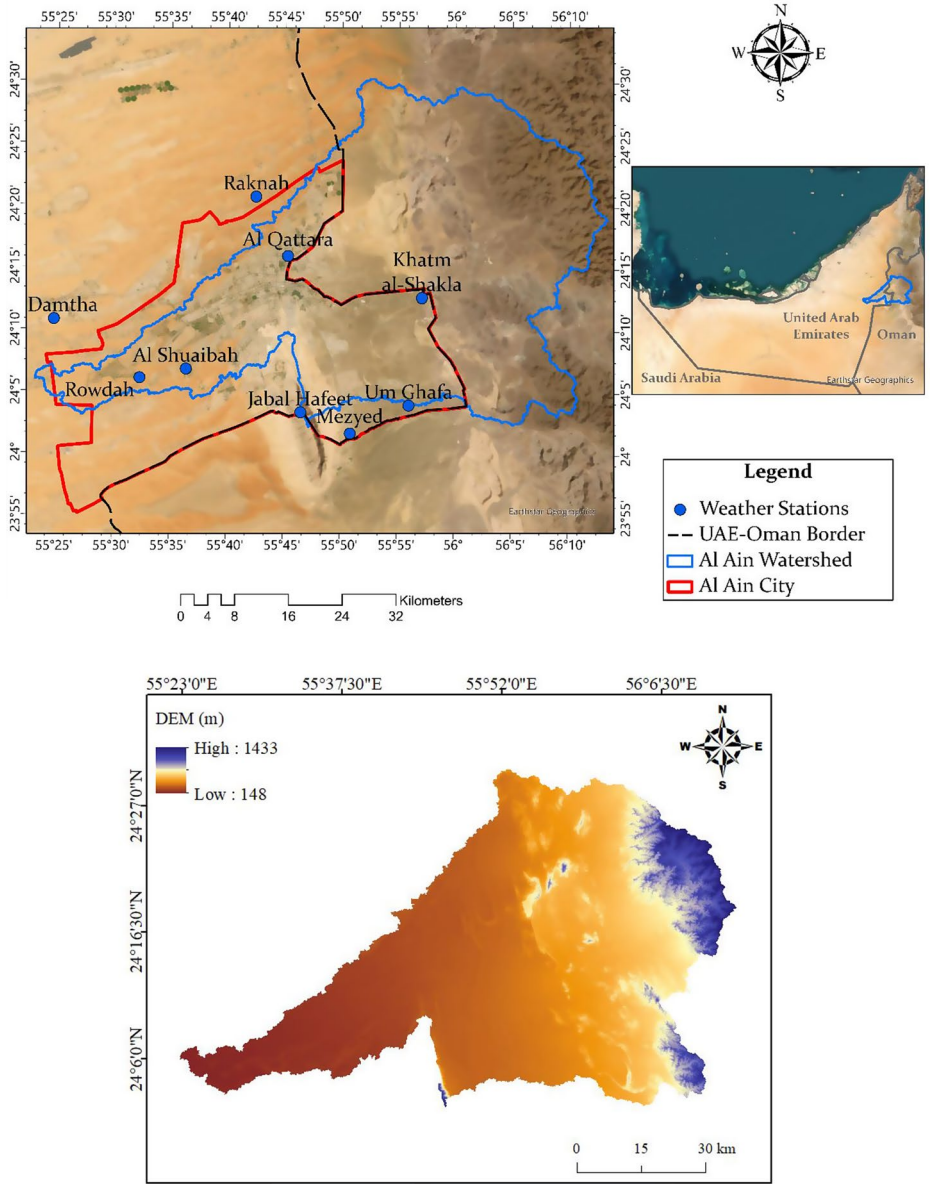


Fig. 3 Location of the Al Ain watershed (top) and its topography (bottom), based on a 12.5 m digital elevation model (DEM)

The Al Ain watershed was delineated using the Advanced Land Observing Satellite Phased Array type L-band Synthetic Aperture Radar (ALOS PALSAR) Global Digital Elevation Model, which has a resolution of 12.5 m. The data was acquired from NASA's Alaska Satellite Facility - Distributed Active Archive Center (ASF), accessible at https://asf.alaska.edu.

2.3 The hydrologic model

The GSSHA model is a physically-based, distributed-parameter tool used to simulate flooding events across catchments of various sizes (Downer and Ogden 2004). Using this model to simulate flooding from the event provides detailed runoff predictions by considering spatial variability and integrating diverse data sources, aiding in effective flood management and risk mitigation by identifying vulnerable areas. The GSSHA model functions as a process-based model, simulating all hydrologic states and fluxes within each grid cell before, during, and after storm events. GSSHA accepts spatially and temporally varying precipitation data from various sources like gauges, radars, satellites, or design storms. It can also account for snowfall accumulation, snowmelt, interception losses, evapotranspiration, surface retention, and infiltration. The model employs a simple two-parameter scheme to simulate plant interception of precipitation.

For infiltration modeling, users can select from four methods: Green-Ampt (GA), Green-Ampt with Redistribution (GAR) (Ogden and Saghafian 1997), multi-layered GA, or the fully implemented Richards' equation (GSSHA Primer 2024). GSSHA goes beyond basic infiltration modeling by offering the GAR method for simulating continuous storm events with intermittent rainfall. Additionally, the model boasts a comprehensive suite of capabilities, including overland runoff routing, unsaturated zone soil moisture dynamics, saturated groundwater flow, surface and in-stream sediment transport and deposition, simplified lake storage and routing, wetland peat layer hydraulics, and overland contaminant transport and uptake (GSSHA Primer 2024).

The GAR method, employed for the April 2024 flood simulation, incorporates inter-storm redistribution of soil water and performs multiple ponding simulations using the GA methodology. The core equation governing this method is:

$$f(t) = K\left(\frac{S_f(\theta_s - \theta_i)}{F(t)} + 1\right) \tag{1}$$

where:

$f(t)$= potential infiltration rate (cm/h)

$F(t)$=cumulative infiltration (cm).

S_f= wetting front suction head (cm).

K=effective hydraulic conductivity (cm/h)=K_s/2.0 under typical field conditions to account for the fact that the saturated hydraulic conductivity (K_s) represents the maximum rate at which water can move through the soil when it is fully saturated.

K_s = saturated hydraulic conductivity (cm/h).

θ_s=water content of the soil at natural saturation.

θ_i=initial soil water content.

The GAR method utilizes the traditional rectangular wetting front assumption for variations of Eq. 1, predicting infiltration across multiple ponding periods. Following the subtraction of abstractions, GSSHA computes ponded flow over each grid cell using the diffusive wave approximation of Saint-Venant's equation. This flow is then routed in two orthogonal directions. For grid cells bordering the watershed divide, only inward flow is permitted (GSSHAwiki 2024). GSSHA offers three numerical schemes to solve the diffusive wave equation: Explicit, Alternative Direction Explicit (ADE), and ADE-Prediction Correction (PC). The physiographic characteristics of the catchment determine the most suitable method. The Explicit scheme is the fastest and simplest, but also the least robust. Conversely, the ADE-PC scheme is the slowest yet most robust (GSSHAwiki 2024). The computationally demanding ADE-PC scheme is generally recommended for complex terrain where minimal smoothing of the digital elevation model (DEM) is desired. This study employs the ADE-PC scheme.

3 Results

The rainfall map (Fig. 4), based on IMERG data, reveals the significant variability in precipitation across the Arabian Peninsula and the UAE during the April 2024 event. The Late IMERG product displays a broader region of intense rainfall, whereas the Early product captures higher localized accumulations, particularly over the coastal areas of the UAE and the mountains of Oman. The most extreme rainfall occurred in the eastern and northeastern parts of Oman and the UAE. Additionally, the map shows precipitation affecting Qatar, Bahrain, Kuwait, parts of Saudi Arabia, and Yemen. In the UAE, heavy precipitation dominated the northern coastal strip near Dubai, Sharjah, and Abu Dhabi, reaching up to 250 mm while some interior regions remained largely unaffected, highlighting the event's localized nature. The data also suggests significant rainfall in Al Ain and the eastern coastal areas. Additionally, there was a pronounced contrast, with the northern Oman Mountains receiving considerably more precipitation compared to the desert regions.

3.1 Rainfall over Al Ain watershed

Table 1 summarizes the total rainfall and maximum 15-minute intensities measured at nine NCM weather stations within and around the Al Ain watershed during the storm event, primarily occurring on April 16th. The table also lists the locations and elevations of these stations. Rowdah, Al Qattara, Raknah, and Khatm al-Shakla stations recorded 24-hour precipitation exceeding the estimated 100-year storm value for the area, as determined by Sherif et al. (2013). Interestingly, the recorded rainfall corresponds to return periods of 750, 1,200, 1,250, and 20,000 years, respectively (Alhosani et al. 2024). The exceptionally high rainfall of 254.8 mm measured at Khatm al-Shakla on April 16th represents 60–75% of the probable maximum precipitation (PMP) estimated by Sherif et al. (2013) using two different methods. Table 1; Fig. 5 show an increase in precipitation with elevation across the Oman Mountains. However, rain gauge data are only available for the portion of the Al Ain watershed within the UAE, covering approximately 50% of the total area. The satellite estimates shown in Fig. 4 align with this orographic precipitation pattern.

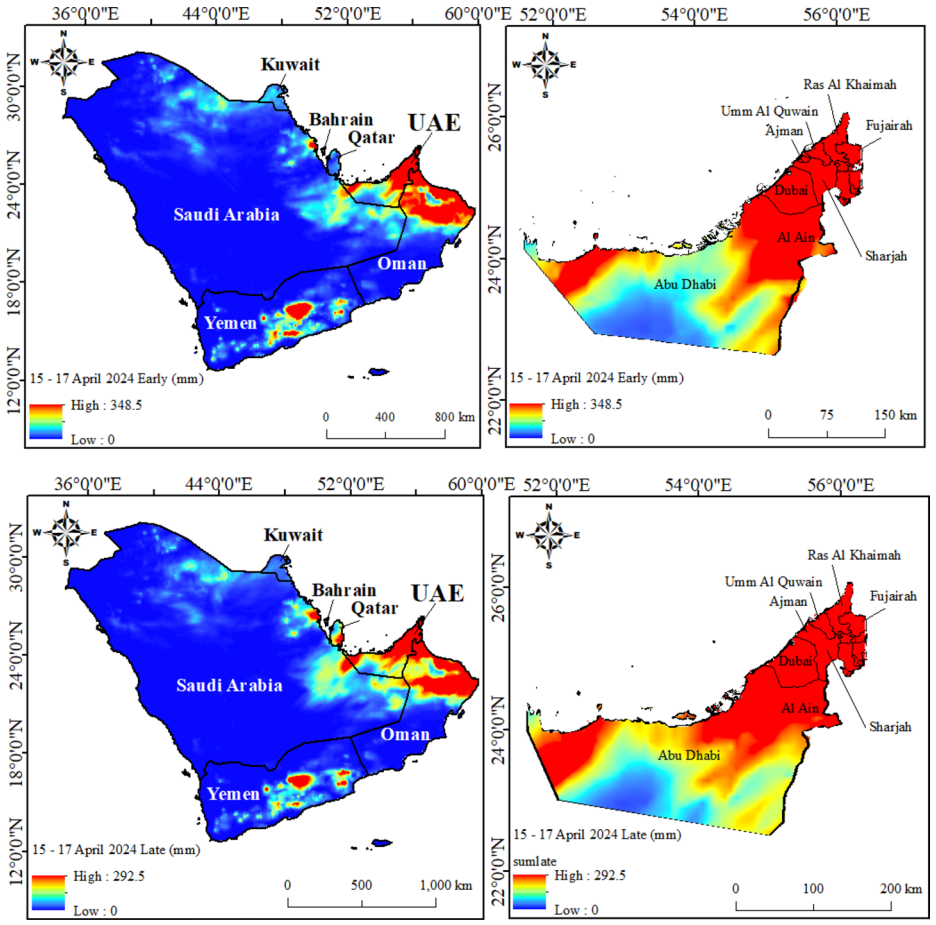


Fig. 4 Total rainfall from IMERG Early and Late estimates over the Arabian Peninsula (left) and the UAE (right) during the April 15–17 storm event. The storm moved from Oman into the UAE on April 16

Table 1 Total rainfall and maximum 15-minute intensity during the April 2024 storm event at 9 weather stations in and around the watershed

Station	Latitude	Longitude	Elevation (m)	Total Rainfall (mm)	Max. 15-min. Intensity (mm/hr.)
Damtha	24.1800000	55.4133333	153	95.62	85.14
Al Shuaibah	24.1138889	55.6072222	215	14.60	2.00
Rowdah	24.1016667	55.5394444	222	173.20	200.80
Al Qattara	24.2661111	55.7552778	280	184.23	159.42
Raknah	24.3455556	55.7080556	288	185.81	138.53
Mezyed	24.0286111	55.8477778	345	83.05	51.50
Um Ghafa	24.0666667	55.9333333	371	91.02	97.03
Khatm al-Shaklah	24.2111111	55.9519444	433	254.80	164.00
Jabal Hafeet	24.0566667	55.7752778	1080	63.20	80.80

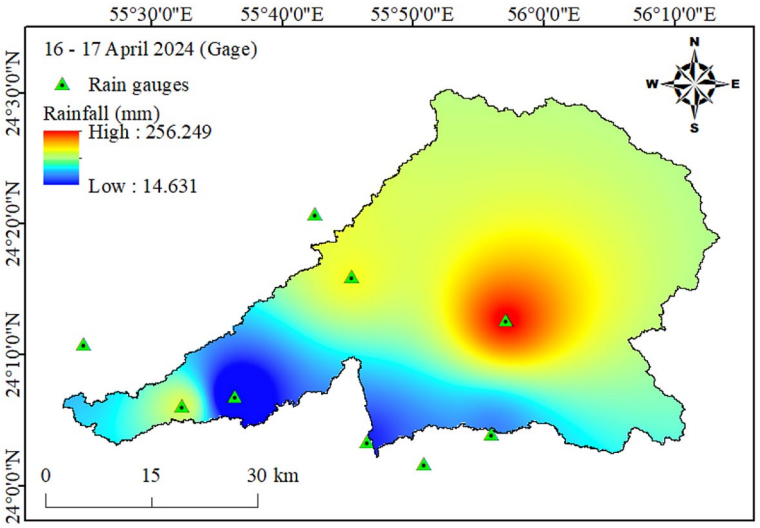


Fig. 5 A map of the interpolated total rainfall for the event over the study watershed, based on observations from the 9 rain gauges

The maximum 15-minute rainfall intensity is a crucial indicator of the severity of short-duration torrential rainfall during a storm. The observed maximum intensity at Raknah station corresponds to a return period of 400 years, while those at Rowdah, Al Qattara, and Khatm al-Shakla stations far exceed the 1,000-year events (Alhosani et al. 2024; Sherif et al., 2013). In contrast, the Al Shuaibah weather station recorded exceptionally low rainfall totals and intensities. The authors suspect that the rain gauge did not operate properly during the event. The Inverse Distance Weighting (IDW) method was used for interpolation in mapping the total rainfall across the Al Ain watershed during the event to illustrate the spatial rainfall pattern (Fig. 5).

Examining the temporal distribution of rainfall during this extreme event provides further insights. This analysis focuses once more on the four stations that recorded the highest rainfall totals. Figure 6 presents hyetographs depicting rainfall at 15-minute intervals for April 16th, covering a 24-hour period. These plots demonstrate significant discrepancies in the temporal distribution and magnitude of rainfall among the stations. However, all hyetographs indicate that the storm comprised three downpour periods, each separated by approximately 6–8 h. Rain began early in the day and continued for about five hours, followed by an eight-hour pause at most stations. A second, more intense episode lasted for about three hours. After another six-hour pause, a final downpour began, with high intensities observed at some stations.

Figure 7 shows the spatial distribution of total rainfall for the event derived from satellite products. Both Early and Late IMERG captured the overall precipitation pattern observed by rain gauges but underestimated the total rainfall compared to ground measurements. The Late product exhibited slightly better agreement with the rain gauge observations (Fig. 5), with the Early IMERG producing the lowest estimates. This contradicts expectations, as climate data adjustments typically reduce estimates in drier regions (e.g., Alsumaiti et al. 2020). However, in this case, one gauge reported an extremely high storm total, while

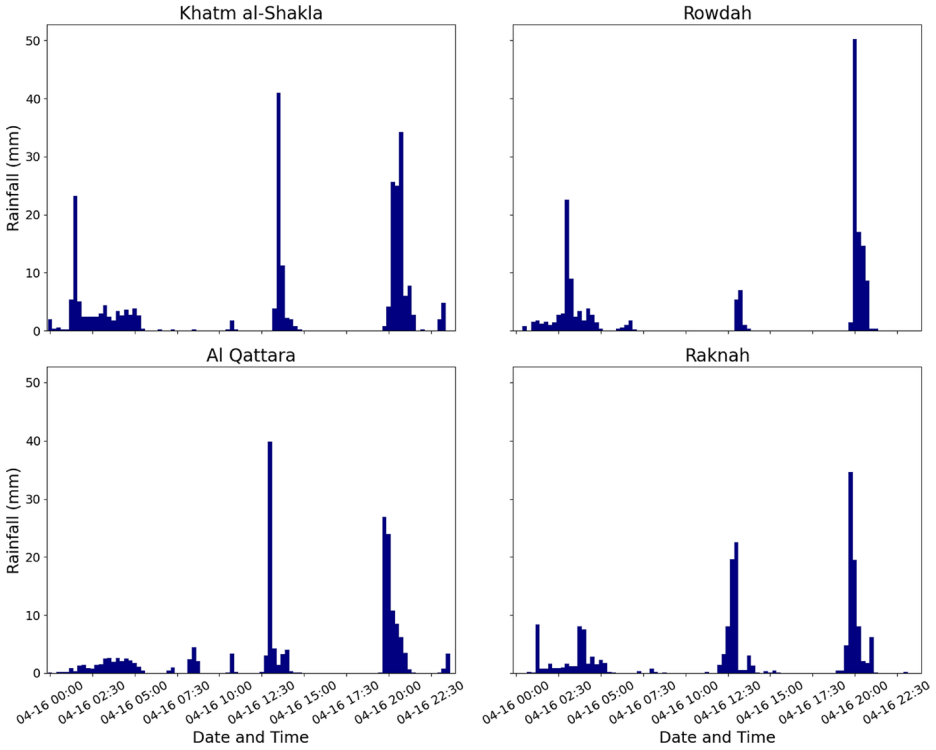


Fig. 6 Hyetographs of 15-minute rainfall at the four weather stations with the highest recorded rainfall accumulations during the storm

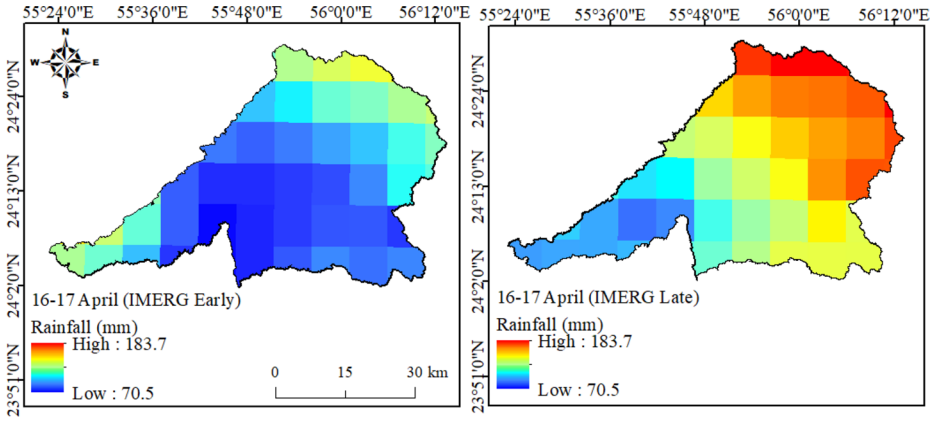


Fig. 7 Rainfall totals for the event as estimated by the IMERG Early and Late run products over the study watershed

another reported a very low estimate. These conflicting observations may have influenced the comparison between the rain gauge and satellite estimates, as well as the adjustments applied to the satellite estimates. Despite the underestimation in total amounts, the temporal patterns of rainfall (variation of precipitation over time) remained relatively consistent across all products.

3.2 Modeling the event

The GSSHA model was used to simulate the flooding caused by the event within the Al Ain watershed. Pre- and post-processing of model inputs and outputs were performed using the Watershed Modeling System (WMS) software (Aquaveo 2024) and ArcGIS. The stream network was refined for accuracy by comparing it to high-resolution imagery of the watershed. To minimize errors, WMS was employed to perform filling and smoothing on a 150 × 150-m grid. This approach simulates all processes within each grid cell, resulting in a fully distributed-parameter hydrologic model.

Soil properties for the Al Ain watershed were obtained from the global digital soil mapping system, SoilGrids™ (https://www.isric.org/explore/soilgrids). This resource provides data on soil texture, including the mass fractions (percentages) of clay, silt, and sand. Using ArcGIS and the soil texture classification system established by the United States Department of Agriculture (USDA) Natural Resources Conservation Service (NRCS) (https://www.nrcs.usda.gov/resources/guides-and-instructions/soil-taxonomy), a soil type map for the watershed was constructed. The developed map was then compared with available high-resolution images to delineate the impervious areas. The final soil type map is presented in Fig. 8. Land use and cover data for the Al Ain watershed was acquired from the OpenLandMap data portal (https://opengeohub.org/about-openlandmap/), with a spatial resolution of 300 m x 300 m. This data was further refined by incorporating information from available high-resolution images. The final land use/cover map is also presented in Fig. 8.

The two-dimensional (2D) overland flow was calculated using the Alternative Direction Implicit (ADE) method, while the one-dimensional (1D) channel flow from each grid cell

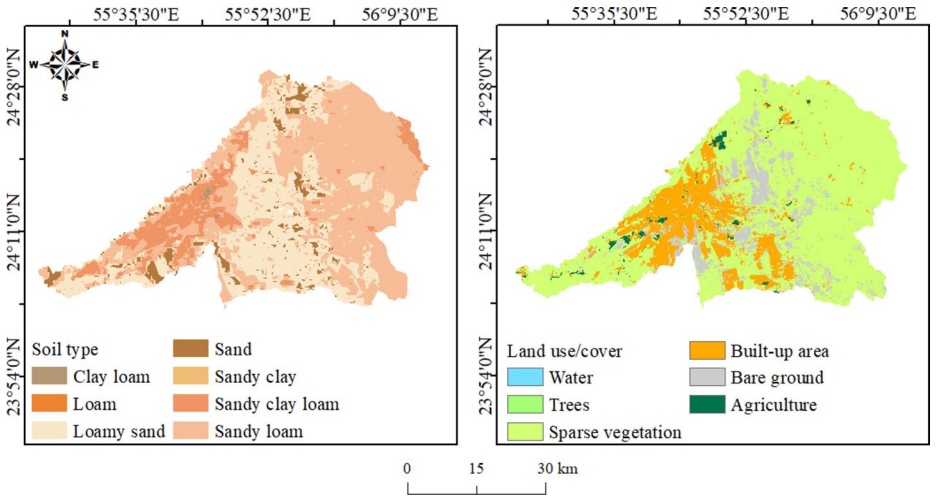


Fig. 8 Maps of soil type and land use/cover for the study watershed

was simulated using diffusive wave equations. Green-Ampt infiltration with redistribution (Ogden and Saghafian 1997) was chosen for infiltration calculations, with hydraulic conductivity values derived from Rawls et al. (1983). Cross-sectional geometries were primarily extracted from the Digital Elevation Model (DEM). Streams with available surveyed data were assumed to have a trapezoidal shape. Manning's roughness coefficients were assigned to land-use/cover classes based on estimations by Sharif et al. (2010a, 2010b). Due to the prevalence of steep slopes, extensive impervious cover, and rapid urban response times, runoff was characterized as primarily Hortonian overland flow (infiltration excess). Consequently, subsurface runoff and groundwater flow were excluded from the model. A roughness coefficient range of 0.02–0.05 was assigned to stream channels.

The storm began on April 14th over Oman and reached the UAE by the late afternoon of the 15th, initially impacting coastal areas. Rainfall intensified around 9 a.m. on the 16th and progressed further inland. By the end of the 16th, Dubai had received over 142 mm of rain within 24 h, significantly exceeding its annual average of 94.7 mm. As the storm continued inland, rainfall peaked during the afternoon and evening. Figure 9 depicts the runoff hydrographs simulated by GSSHA at the watershed outlet. Variations in peak flow and hydrograph duration reflect the differences in rainfall intensity and distribution (Fig. 6). Remarkably, the resulting runoff represents a small fraction of the total rainfall, approximately 6.4% or less (Table 2). This low runoff can be attributed to the watershed's characteristics, total rainfall volume, and the rainfall's spatial and temporal variability. The hydrographs' relatively flat shapes and sudden slope changes reflect the temporal pattern of the rainfall. While the peak discharge is significant and likely unprecedented, it could have been much larger if the rainfall had been a single downpour instead of three separate events with 6–8-hour intervals. While the simulated runoff hydrographs driven by satellite rainfall data captured the general pattern of those derived from rain gauges, they also amplified the rainfall estimation errors (Fig. 9; Table 2). This highlights how inaccuracies in rainfall estimates can significantly magnify or diminish predicted urban flood severity.

One of the primary strengths of the GSSHA model is its ability to simulate spatially varying processes. This capability is illustrated in Fig. 10, which shows the maximum flood

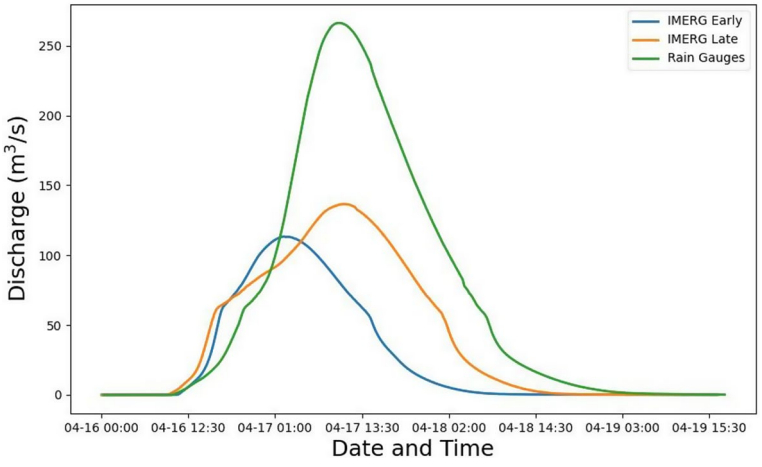


Fig. 9 Hydrograph simulated by GSSHA at the outlet of the study watershed when the model was driven by rain gauge observations and satellite estimates

Table 2 Rainfall volume and GSSHA-simulated runoff ratio and percentage and peak discharge driven by rain gauge and satellite data

Rainfall data	Rainfall volume (m^3)	Runoff volume (m^3)	Runoff ratio (%)	Peak discharge (m^3/s)
Rain gauges	343989924.4	21993129.8	6.4	266.3
Early	213311258.5	08341843.0	3.9	113.6
Late	304827171.8	13146159.2	4.3	136.5

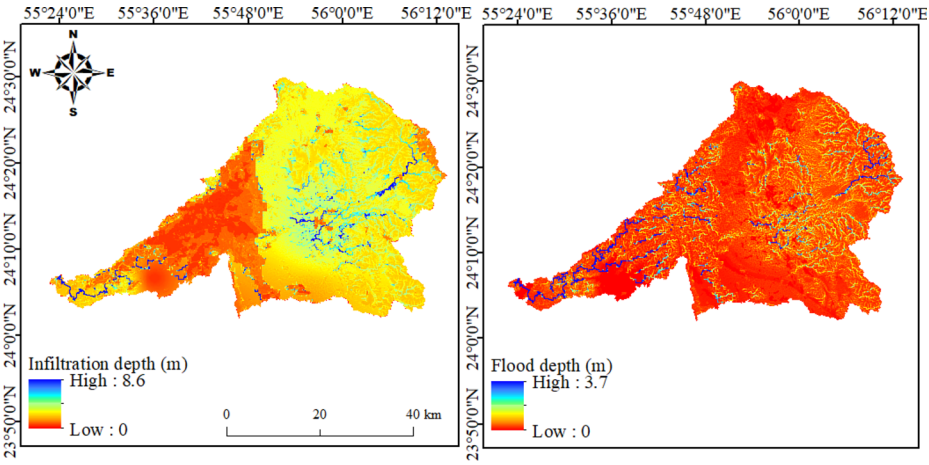


Fig. 10 Maps of the total cumulative infiltration depth (left) and maximum surface runoff depth (right), as predicted by GSSHA when driven by rain gauge observation

depths and total infiltration depths within the Al Ain watershed during the event. The model identifies several urban areas and major roads at risk of flooding, a risk linked to historical development trends in arid regions where communities often develop around wadis that later become main transportation routes. These locations are particularly susceptible to runoff accumulation due to their low infiltration rates. It is important to note that the simulated flood extent and depths are influenced by the underestimation of rainfall from satellite data compared to rain gauge measurements.

4 Discussion and conclusions

In April 2024, a storm event across the Arabian Peninsula and the UAE displayed significant variability in rainfall patterns, ranging from localized accumulations to widespread, intense rainfall areas. Mountainous regions are particularly prone to intense rainfall and flash floods due to orographic precipitation, steep slopes, and limited infiltration. The Oman Mountains, a significant geological feature, increase rainfall in the Al Ain region, heightening the risk of flash floods. Populations in these areas, often rural, rely on natural resources and are highly vulnerable to extreme weather events.

Integrating satellite-derived data, rain gauge observations, and hydrological modeling provides a comprehensive understanding of how extreme weather events influence water-

related challenges and infrastructure resilience in arid environments. This knowledge is crucial for enhancing forecasting accuracy and preparedness measures to mitigate future storm impacts in the region. However, IMERG rainfall products often underestimate ground measurements in the UAE, likely due to the region's unique climatic conditions, such as high humidity, dust storms, and complex topography, which can interfere with satellite data accuracy (Alhosani et al. 2024; Lu et al. 2021; Alsumaiti et al. 2020). Moreover, the sparse distribution of ground-based rain gauges in the UAE limits the calibration and validation of satellite estimates. Potential solutions include enhancing the density and distribution of ground-based measurement stations, improving satellite algorithms to better account for regional climatic factors, and integrating data from multiple satellite sources to increase accuracy and reliability.

The April 2024 event highlights the increasing frequency of flash floods in arid regions like the Arabian Peninsula, emphasizing the necessity for a deeper understanding of these processes. Assessing current infrastructure capacity to manage future risks is particularly critical (Rözer et al. 2023). Research indicates that runoff potential during such events depends not only on storm characteristics but also on environmental factors, with urbanization notably exacerbating flood risks (Shawky and Hassan 2023). Further investigation into these phenomena is essential to improve preparedness and mitigate the destructive impacts of flash floods in these regions (e.g., Kurata et al. 2023; Nakhaei et al. 2023; Nagamani et al. 2024). Ramadan et al. (2022) argue that understanding specific local factors contributing to flash floods can significantly enhance early warning systems and disaster response strategies.

Addressing the challenges of complex terrain for rain gauge placement can involve using remote sensing technologies, drones, and community-based monitoring. These strategies enhance rainfall monitoring and early warning systems, reducing risks to vulnerable populations in these regions. Furthermore, while traditional stream gauges struggle to accurately capture pluvial flooding in urban environments, advancements in low-cost, high-tech sensors (e.g., Mydlarz et al. 2024) offer a promising solution. These sensors provide real-time data crucial for emergency response efforts, enabling agencies to target rescue operations, strategically barricade roads, and deliver precise impact-based forecasts. Additionally, leveraging airborne LiDAR and terrestrial LiDAR technologies is highly recommended for creating detailed Digital Surface Models (DSMs) and Digital Terrain Models (DTMs) of Al Ain city. Advanced LiDAR technologies offer high-resolution elevation data essential for robust flood modeling and infrastructure resilience planning tailored to Al Ain's unique topography.

One crucial element in preparedness is impact-based forecasting (IBF). This approach goes beyond simply predicting the weather (Harrowsmith et al. 2020; Robbins et al. 2022a, b). IBF takes into account factors like population density, infrastructure vulnerability, and the potential consequences of the event. With this information, forecasters can provide targeted warnings that focus on specific risks to different areas and populations (Harrowsmith et al. 2020; Robbins et al. 2022a, b). By developing IBF capabilities, the country can move towards forecasts that predict not just the weather but also the potential impact on communities. This requires investment in data collection on infrastructure, population density, and vulnerability in different areas. Additionally, strengthening communication channels is vital to ensuring targeted warnings reach the public effectively. By implementing IBF, the UAE can better prepare for future extreme weather events and minimize the associated risks.

The UAE is actively developing resilient urban areas in response to climate change. Integrating climate considerations into planning frameworks, exemplified by Abu Dhabi's 2030 Urban Structure Framework Plan, underscores this commitment (Plan 2030, 2024). Encouraging forward-thinking initiatives that prioritize adaptive and future-oriented urban planning practices is crucial (EEA, 2012; UN, 2019; Abdelkareem and Mansour 2023). Key focuses include incorporating climate change projections, enhancing infrastructure resilience, and aligning land use with water management strategies. To mitigate significant flood risks in low-lying coastal regions and residential areas near wadis, several measures can be implemented. This includes enhanced urban planning and stricter zoning laws to restrict construction in high-risk zones, alongside investments in robust flood defense systems and improved drainage infrastructure, all of which are essential (DOLA, 2020; FEMA, 2021; Sofia et al. 2023). Nature-based solutions such as wadi rejuvenation and the creation/restoration of natural wetlands, coupled with integrated water management strategies, can effectively manage excess water (Cohen-Shacham et al. 2016; Kabisch et al., 2017). Establishing early warning systems and promoting climate-resilient building practices are crucial for preparedness and resilience (UNDRR, 2020). Engaging communities through education and strengthening policies at both national and local levels will further support efforts to protect vulnerable populations from flooding (DOLA, 2020; IPCC, 2022).

The UAE experiences relatively few flood fatalities compared to other affected countries. Research into the reasons for this lower fatality rate could yield valuable insights. Understanding these factors can inform best practices and strategies to minimize flood-related deaths. Implementing and improving proactive measures based on these insights can further reduce fatalities in future flood events. By learning from past successes and continuously improving safety protocols, the UAE can ensure even greater protection for its residents against the threat of floods. Interestingly, all five individuals who died in the recent floods in the UAE were in vehicles at the time. This underscores the critical need for improved public awareness and warning systems about the dangers of driving in flood conditions. This tragic outcome is not unique to the UAE; globally, many flood-related fatalities occur in vehicles (e.g., Jonkman and Kelman 2005; Kellar and Schmidlin 2012; Han and Sharif 2020). There are several scientific and practical reasons for this, including low visibility, underestimating the power of floodwaters, and ignoring flood warnings, as well as strategies to mitigate such risks (e.g., Kellar and Schmidlin 2012; IPCC 2014; Han and Sharif 2021).

Despite the extreme aridity of the Arabian Peninsula, flash floods from sudden downpours pose an increasing threat, particularly to rapidly developing urban centers near impermanent wadis (dry stream beds). A stark illustration of this danger occurred across large areas of the UAE in mid-April 2024. Unprecedented torrential storms resulted in rainfall equivalent to two years' worth at some weather stations, shattering all previous records. For instance, Khatm al-Shakla in the Oman Mountains recorded an astonishing 254.8 mm in a single day, with rainfall intensity peaking at an alarming 200.8 mm/hour over a 15-minute period at another station. Although the resulting runoff caused property damage, flooded major streets, and tragically claimed five lives, coordinated crisis management efforts averted a full-blown disaster.

These events highlight the critical role of fully-distributed, physically-based hydrologic models in providing a detailed depiction of runoff and flooding dynamics in densely urbanized watersheds. Such models, when combined with enhanced rainfall forecasts, become invaluable tools for predicting flood extent and impact. They also aid in identifying at-

risk roads, which is crucial for preventing flood-related fatalities where drivers encounter flooded areas. This capability allows authorities to issue targeted warnings and implement proactive measures like barricading flood-prone areas. Furthermore, emerging distributed sensing techniques for urban flooding show significant promise for calibrating and validating flood models in arid cities, overcoming the limitations of traditional methods for measuring channel discharge.

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Authors contributions K.H and N.A. guided this research and contributed significantly to preparing the manuscript for publication. K.H., N.A., and H.S. developed the research methodology. M.A., A.M.A, and A.A. downloaded and processed the remote sensing products. A.M.A, and A.A. developed the scripts used in the data analysis. K.H., A.M.A, A.A. and H.S. performed the modelling and analysis of the results. K.H., N.A., T.A., H.S., A.M.A, A.A., M.A, A.M.G.A. and W.A. prepared the first draft. K.H., N.A., W.A. and H.S. performed the final overall proofreading of the manuscript.

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Declarations

Competing interests The authors declare no conflicts of interest. The funding sources was not involved in the study's design; in the data collection, analysis, or interpretation; in the manuscript writing; or in the decision to publish the results.

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Authors and Affiliations

Khalid Hussein[1,2] · Naeema Alhosani[1] · Ahmed M. Al-Areeq[3,4] · Amran A. Al Aghbari[5] · Muhagir Elkamali[1] · Tareefa Alsumaiti[1] · Hatim O. Sharif[6] · Ahmed M. G. Almurshidi[1] · Waleed Abdalati[2]

✉ Khalid Hussein
khalid.hussein@uaeu.ac.ae

1 Geography and Urban Sustainability Department, College of Humanities and Social Sciences, United Arab Emirates University, Al Ain, UAE

2 Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado, Boulder, CO 80309, USA

3 Department of Civil and Environmental Engineering, King Fahd University of Petroleum and Minerals (KFUPM), Dhahran 31261, Saudi Arabia

4 Interdisciplinary Research Center for Membranes and Water Security (IRC-MWS), King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia

5 College of Engineering and Physics, King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia

6 School of Civil & Environmental Engineering and Construction Management, University of Texas at San Antonio, San Antonio, TX 78249, USA